Tracking structural cues or relying on probabilistic inference in Turkish agreement?

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Agreement attraction occurs when a plural attractor makes an ungrammatical sentence appear acceptable, as in '**The key to* <u>*the cabinets*</u> *are rusty*.' What factors drive such errors in agreement comprehension? Retrieval-based accounts attribute those to occasional erroneous retrieval of the plural attractor from memory due to either its surface features matching those of the verb like number or case[1-2] or the association of these cues with an abstract notion of agreement controller like subjecthood.[3-6] An alternative account posits that comprehenders rely on probabilistic inferences on the strings they read and may perform a repair of the string into a more probable, well-formed alternative, such as assuming a plural target as in '<u>*The keys*</u> *to the cabinets are rusty.*'[7-8] Here we compare the predictions of these accounts using Turkish possessive constructions featuring GEN(itive) vs. NOM(inative)-marked (bare) attractors. While both attractors can surface as subjects and thus are associated with agreement controllerhood in Turkish,[4,9] only bare attractors match the surface features of a tensed verb. Conversely, corpus data show that GEN-marked attractors generate higher probabilities for a grammatical, plural target than bare attractors. Through an acceptability task we show evidence for comparable attraction errors with both attractors, suggesting that <u>abstract structural cues associated with subjecthood, rather than surface cue matching or probabilistic repair, plays a dominant role in driving attraction effects.</u>

EXPERIMENT. A speeded, binary acceptability task ($N_{participant}$=59, N_{item}=24) compared agreement errors with two attractor types in a 2x3 within-subject design, crossing attractor CASE (GEN vs. bare) and AGREEMENT MATCH (target, attractor or none) (see 1). The verb was always plural marked. Only the sentences with a matching (plural) target were grammatical.

PREDICTIONS. <u>Retrieval accounts</u> that attribute agreement errors to the attractor's abstract structural role predict comparable attraction effects with both attractors, as both are potential structural controllers in the language. However, those that focus on the match of surface features (e.g., number, case) with retrieval cues predict higher attraction with bare attractors, as they are more similar to the target in the sentence. To index this association in a large language model, we calculated attention weights directed toward the target vs. attractor using BERT, a transformer:[10-11] This analysis (**Fig3**) showed increased attention to bare attractors when they are plural compared to singular, but no such reliable difference with GEN-marked attractors. Under <u>probabilistic inference accounts</u>, comprehension involves a repair mechanism where listeners consider the likelihood of alternative strings of the perceived input due to potential noise.[12]Attraction should thus be modulated by the frequency of the alternative form of the target; with increased errors when the prior probability of the target being plural is higher with an attractor. We used **CORPUS ANALYSES** to calculate the conditional probability of the target's number given the attractor's (see Table 1): With GEN-marked attractors, a plural target is more likely with a plural than a singular attractor (24% vs. 11%); With bare attractors, the pattern reverses: a plural target is less likely with a plural than a singular attractor (5% vs. 17%). This predicts attraction with plural GEN-marked attractors but weak(er) or lack of attraction with plural bare attractors.

RESULTS. For both CASE markings, participants gave more 'yes' responses in ATTRACTOR MATCH (M_{GEN}=34.3, M_{BARE}=28.4) than NO MATCH conditions (M_{GEN}=29.2, M_{BARE}=18.6) (**Fig1**). A crossed Bayesian GLM model (**Fig2**) verified this: There was strong evidence for an overall difference between ATTRACTOR and NO MATCH (β = 0.26 [.05,.48], P(β>0) = 0.99), but this did not interact with CASE (β = -0.05 [-.29,.20], P(β>0) = 0.34). A nested Bayesian GLM model showed reliable ATTRACTOR vs. NO MATCH differences within each attractor type (GEN: β = 0.20 [-.10,.51], P(β>0) = 0.91; bare: β = 0.31 [-.01,.64], P(β>0) = 0.97).

OVERALL, we found similar rates of attraction errors with two differently case marked attractors, which are both controller-like but differ in their co-occurrence probabilities with grammatical target nouns and their surface cue matching. This suggests evidence that abstract structural cues related to subjecthood is an important determining factor in agreement errors.

Stimuli. The versions with *–(n)In* show the Genitive conditions, and those without show the Nominative (bare) conditions. Agreement target is underlined; and attractor is bolded.

a. TARGET MATCH	b. ATTRACTOR MATCH	c. NO MATCH
Mülteci(-nin) <u>avukat-lar-ı</u>	****Mülteci-ler(-in)*** <u>avukat-ı</u>	****Mülteci(-nin)*** <u>avukat-ı</u>
refugee(-GEN) lawyer-PL-POSS	refugee-PL(-GEN) lawyer-POSS	refugee(-GEN) lawyer-POSS

… duruşmada durmadan bağır-dı-lar.
… at.hearing non-stop scream-PST-PL
'The refugee(s)'s lawyer(s) shouted at the hearing non-stop.'

Attractor GEN	Target	Counts (out of 1m)	Pr(TargetPL | Att)	Attractor BARE	Target	Counts (out of 1m)	Pr(TargetPL | Att)
SG	SG	4901.8	11.4%	**SG**	SG	18407	**17%**
	PL	631.5			PL	3842.4	
PL	SG	1684.7	**24%**	PL	SG	655	5%
	PL	540.3			PL	37	

Table 1. Counts for each combination of number features in [attractor, target] tokens in the TrTenTen corpus (~4.9 billion words). Critical rational Inference predictions are highlighted.

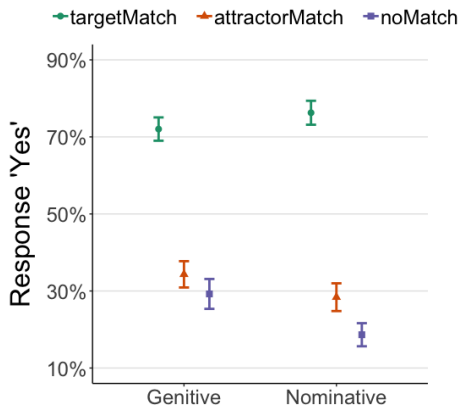

Fig1. Mean percentages of 'Yes' responses in the judgment task. Error bars show by-participant SEs.

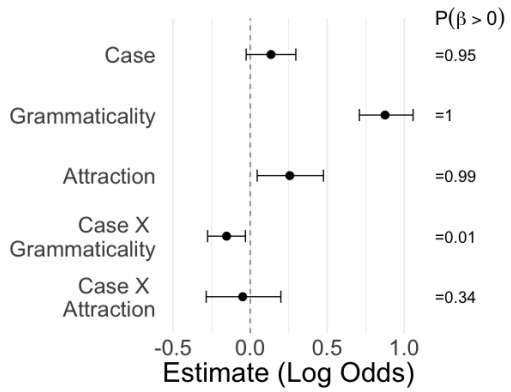

Fig2. Posterior summaries (mean, 95% CrI, and P(β>0)) from the Bayesian GLM model. Case = GEN vs. bare; Grammaticality = targetMatch vs. attractorMatch/noMatch; Attraction = attractorMatch vs. noMatch.

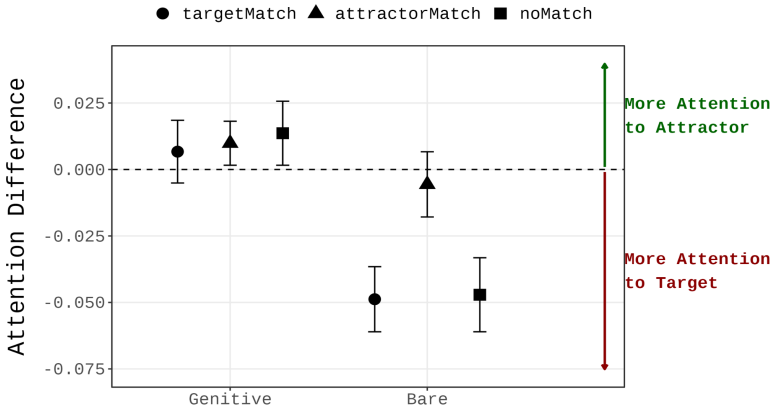

Fig3. Attention as a correlate of high-level controllerhood abstraction. Y-axis shows the difference between Attention from verb to attractor vs. target.

References:

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[2] Lewis & Vasishth, 2005
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[8] Cartner et al., 2025
[9] Göksel & Kerslake, 2005
[10] Ryu & Lewis, 2025
[11] Viota et al., 2019
[12] Levy, 2008